

LINKÖPING MALMEN, Sweden

WMO: 025620

Lat: 58.402N

Lon: 15.526E

Elev: 308

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.2	(c) 8.9	(d) -1.1	(e) 5.2	(f) 2.9	(g) 5.2	(h) 7.3	(i) 10.5	(j) 27.3	(k) 41.3	(l) 24.5	(m) 41.4	(n) 2.6	(o) 250	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	80.4	65.1	77.1	64.0	73.7	62.3	67.6	75.8	65.5	73.6	63.8	71.3	7.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	92.6	70.6	62.7	86.0	68.8	60.7	80.3	67.1	32.1	76.3	30.5	73.7	29.2	71.3	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	19.8	17.5	-3.4	84.6	7.8	4.2	-9.0	87.6	-13.6	90.0	-17.9	92.3	-23.6	95.4
			-3.9	70.3	7.7	2.3	-9.4	72.0	-13.9	73.3	-18.2	74.6	-23.8	76.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	28.3	28.4	33.4	42.3	50.7	57.7	62.4	61.0	53.5	44.5	36.8	30.9
	DBStd	14.00	8.90	9.13	7.67	6.39	6.37	5.05	5.15	4.90	5.48	6.42	7.02	9.14
	HDD50	3313	672	604	514	244	68	3	0	0	28	191	397	592
	HDD65	7640	1137	1024	979	682	445	227	117	144	345	637	846	1057
	CDD50	1211	0	0	0	12	89	233	383	340	134	19	1	0
	CDD65	62	0	0	0	0	1	7	35	19	0	0	0	0
	CDH74	663	0	0	0	1	26	104	372	156	4	0	0	0
	CDH80	106	0	0	0	0	1	11	74	20	0	0	0	0
Wind	WSAvg	8.2	8.5	8.9	8.9	8.1	7.8	7.9	7.2	7.2	8.0	8.3	8.3	8.9
Precipitation	PrecAvg	21.7	1.5	1.0	1.1	1.3	1.5	1.5	2.8	3.0	2.2	1.8	2.3	1.7
	PrecMax	28.0	2.8	2.9	2.3	2.8	3.4	2.8	4.4	5.9	4.3	4.7	4.7	4.7
	PrecMin	15.5	0.3	0.4	0.0	0.0	0.4	0.4	0.9	0.6	0.7	0.2	0.9	0.3
	PrecStd	3.8	0.7	0.7	0.7	0.7	1.0	0.6	0.9	1.7	1.0	1.2	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	48.4	56.5	68.4	77.4	81.5	86.2	82.2	73.8	62.0	54.0	51.0
		MCWB	44.9	45.2	46.8	54.8	62.3	64.6	67.0	65.2	63.0	56.9	51.4	48.7
	2%	DB	43.6	44.6	51.4	63.0	72.1	77.1	81.8	78.3	69.3	58.4	51.1	46.9
		MCWB	41.6	42.0	44.3	51.4	58.0	63.2	66.0	64.7	60.6	54.9	48.9	45.0
	5%	DB	41.1	42.4	47.4	58.3	68.0	73.0	78.2	74.9	65.7	56.0	48.9	44.6
		MCWB	39.3	40.2	42.2	49.1	56.7	60.7	64.6	63.5	59.0	53.2	47.1	42.9
	10%	DB	39.1	40.1	44.2	54.4	63.7	69.1	74.6	71.3	62.7	53.8	46.5	42.2
		MCWB	37.4	38.0	40.2	46.5	53.8	58.7	63.4	62.1	57.1	51.3	45.0	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.2	45.2	48.4	56.9	64.3	67.3	70.5	69.4	65.0	58.2	52.2	49.2
		MCDB	46.6	47.4	54.6	66.2	74.8	77.1	80.8	77.1	71.0	61.0	53.5	50.8
	2%	WB	42.0	42.5	44.9	52.5	60.6	64.2	68.3	67.0	61.8	55.6	49.1	45.4
		MCDB	43.2	44.3	49.7	59.9	68.6	73.3	77.6	73.4	67.5	57.7	50.5	46.7
	5%	WB	39.5	40.3	42.8	49.8	57.6	62.0	66.2	65.0	59.7	53.6	47.2	43.1
		MCDB	40.8	42.2	46.5	57.0	65.5	70.8	75.2	71.5	64.5	55.6	48.7	44.4
	10%	WB	37.5	38.1	40.6	47.3	55.0	60.0	64.3	63.3	57.8	51.7	45.0	40.7
		MCDB	38.8	39.8	43.9	53.3	62.7	67.6	72.4	69.9	61.8	53.6	46.2	41.9
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.7	12.9	17.0	18.3	17.5	17.5	16.5	14.5	11.2	7.9	8.3
		MCDBR	9.0	9.5	17.7	24.8	26.2	24.9	24.7	22.0	19.4	12.6	9.8	9.3
		MCWBR	8.2	7.9	11.9	14.8	14.4	12.8	11.1	10.6	11.5	9.7	8.8	8.8
	5% WB	MCDBR	8.9	9.6	16.4	21.8	22.0	21.4	21.5	18.3	17.5	11.9	9.6	9.3
MCWBR		8.2	8.3	11.7	13.7	13.1	12.1	10.9	10.1	11.2	9.7	8.8	8.8	
Clear-Sky Solar Irradiance	taub	0.279	0.317	0.336	0.374	0.370	0.359	0.385	0.376	0.345	0.335	0.306	0.266	
	taud	2.239	2.299	2.358	2.318	2.392	2.445	2.393	2.414	2.487	2.429	2.299	2.176	
	Ebn at Noon	177	225	257	264	274	279	267	261	253	218	165	139	
	Edh at Noon	15	22	28	34	34	33	34	31	25	20	14	11	
All-Sky Solar Radiation	RadAvg	111	307	787	1267	1638	1824	1633	1299	880	400	143	77	
	RadStd	16	51	88	144	144	87	192	136	121	66	27	8	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	-268	N/A	N/A
(47) Regional (56 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	-236	-284	N/A	N/A

Nomenclature: See separate page